The Impact of Climate Change on Quebec's Tourism Industry

Introduction

Climate change is a pressing global concern with far-reaching implications, and one of its significant yet often overlooked impacts is on the tourism industry. This paper aims to shed light on the profound effects of climate change on tourism, with a specific focus on Quebec, Canada. Understanding these effects is crucial, not only due to the importance of tourism for Quebec's economy, but also because it serves as a microcosm of the broader challenges faced by the global tourism sector in the era of climate change. Quebec, with its diverse natural attractions and unique culture, has long been a magnet for tourists. However, the province's susceptibility to climate change poses a substantial risk to its tourism industry. This paper is structured to provide a comprehensive understanding of the multifaceted relationship between climate change and tourism in Quebec. It will delve into the specific consequences of climate change on Quebec's tourism, explore the adaptation strategies being implemented within Quebec's tourism sector, and examine the economic consequences of climate change on Quebec's tourism. The final section of this paper will present a future outlook and recommendations based on findings. Through recommendations for policymakers, businesses, and stakeholders, this paper aims to contribute to a sustainable and resilient future for Quebec's tourism sector in the face of the ongoing climate crisis.

Climate Change and Tourism in Canada

Climate change is an overarching global issue that significantly affects various sectors, and the Canadian tourism industry is no exception. As a vast and diverse country, Canada's tourism industry thrives on its reputation as a land of pristine wilderness, iconic national parks, and unique wildlife. However, these assets are under threat as climate change alters the country's environmental conditions. The consequences are multifaceted and encompass a range of key challenges and trends with nationwide ramifications. The economy, in particular, faces considerable risks as the tourism sector grapples with climate-related disruptions. Climate change-induced shifts in the migration and breeding patterns of wildlife, the lengthening of certain seasons and the unpredictability of weather patterns, and the gradual loss of glaciers and diminished snowfall in various regions all have dire consequences for the tourism industry in Canada (Environment and Climate Change Canada, 2022). Additionally, the increasing frequency and severity of extreme weather events, such as wildfires and floods, disrupt travel plans, damage infrastructure, and raise safety concerns for tourists. These events can result in direct economic losses for businesses in affected areas. Coastal regions in particular are increasingly vulnerable to sea-level rise and storm surges ("How climate change affects coastal regions," 2021). Pedro Antunes, chief economist at the Conference Board of Canada, says that although "on a national basis, tourism isn't as vital to the country as some Caribbean nations, for example, it can have an outsized impact on some resort towns in Canada like the ones that are affected by the B.C. travel ban" (Lord, 2023). The economic implications of climate change on Canadian tourism are substantial, as the sector contributes approximately thirty-eight billion dollars to the nation's GDP, and supports 1.9 million jobs ("Government of Canada releases new Federal Tourism Growth Strategy," 2023). Because of this, the sector's adaptability and resilience in the face of these challenges are crucial for maintaining economic stability in affected regions.

Quebec's Unique Position

Quebec, known for its rich cultural heritage, stunning landscapes, and vibrant cities, holds a significant position as a prime tourist destination within Canada. Quebec is one of the most popular tourist destinations in Canada. The province's tourism industry also contributes to its economy—according to the Quebec Ministry of Tourism, in 2021, tourism accounted for approximately 1% of Quebec's GDP, generating around $7 billion in revenue. This number was more than double in previous years, but the COVID-19 pandemic prohibited travel into and out of the province, resulting in diminishing numbers.

The province's diverse attractions, including vibrant cities like Montreal and Quebec City, historic sites, natural wonders such as the Laurentian Mountains (a popular destination for alpine skiing), and unique cultural experiences draw tourists from across the country and around the world. Quebec consistently ranks among the top provinces visited by international travellers ("Canada tourism: Trips by province 2022," 2023). The province is also a hub for language and cultural tourism, with many visitors coming to improve their French language skills and immerse themselves in the French-speaking culture. Montreal, in particular, is known for its language schools and cultural diversity ("Montreal: City of diversity and inclusivity," 2021).

However, the province's geographical and climatic characteristics make it susceptible to the effects of climate change, impacting its tourism industry in various ways. Quebec's Far North, where permafrost has historically remained frozen year-round, is experiencing rapid thawing due to rising temperatures. This phenomenon threatens tourism in the region by causing infrastructure damage, altering the stability of buildings, roads, and runways, which in turn limits access to the area due to unstable ground. In addition to this, more densely populated cities like Montreal and Quebec City, located in southern Quebec, are increasingly vulnerable to extreme climate events, including heatwaves, heavy rains, violent winds, and thunderstorms. These events pose challenges for planning outdoor activities such as festivals, impacting the health and safety of visitors. Another notable example is that the regions along the St. Lawrence River, such as Bas-Saint-Laurent, Côte-Nord, Gaspésie, and

Îles-de-la-Madeleine, are facing increased erosion and coastal flooding due to climate change ("Tourism," n.d.). These areas are popular for maritime and coastal tourism, attracting both residents of Quebec and tourists from outside the province. However, climate change-related environmental degradation and infrastructure damage could compromise the appeal of these destinations, potentially leading to a decline in visitor numbers.

The Impact of Climate Change on Quebec's Tourism

Climate change is exerting a substantial impact on Quebec's tourism industry across various facets, leading to disruptions in traditional outdoor activities, infrastructure damage, health hazards, and alterations in wildlife behaviour. These consequences significantly influence the overall tourism experience in Quebec.

One prominent effect of climate change is the alteration of weather patterns. Quebec has experienced extreme weather events such as droughts, heatwaves, heavy rain, and violent winds, as outlined in the Quebec government's climate change impact report ("Climate change impacts," n.d.). These events disrupt outdoor tourism activities and pose significant health and safety hazards for tourists. Heatwaves, in particular, have been linked to respiratory issues, and the warmer temperatures have contributed to the emergence of zoonotic infectious diseases like Lyme disease.

The impact on infrastructure is another notable concern. Freeze-thaw cycles, a consequence of temperature fluctuations driven by climate change, have resulted in damage to roads and other critical infrastructure in Quebec ("Climate change impacts," n.d.). This, along with the increase in torrential rain events, which erode mountain hiking trails, not only hampers the movement of tourists but also affects the accessibility of tourist destinations. Additionally, heavy rains have caused sewage system overflows and increased flood risks, while rising sea levels threaten coastal areas along the St. Lawrence River, further impacting tourism (McWilliams, 2022).

The winter tourism sector in Quebec faces significant challenges due to climate change. Ski resorts have been hit hard as a result of delayed snowfall, shorter snow cover duration, and rising temperatures. An article from Ouranos states that “in northern regions, warming is more pronounced than elsewhere in the world. While the Earth has warmed by 0.8°C over the last 70 years, Quebec’s average annual temperature has increased by 1.1°C” ("Understanding climate change," n.d.). This trend makes skiing an increasingly exclusive and expensive sport, endangering the $295 million skiing-related revenues recorded in Quebec in 2016-2017. It has been predicted that most ski resorts will not be viable by 2080, as the statistics are proving that the skiing season is becoming shorter and less profitable (McWilliams, 2022). Moreover, ice fishing has become less predictable and less profitable due to warm weather and rain, which have rendered the ice unstable. Outfitters on the Sainte-Anne River estimate potential losses exceeding $1 million due to delayed ice fishing seasons ("Warm weather and rain dampen winter sport enthusiasm," 2023).

The St. Lawrence River, a central element of Quebec's tourism landscape, has witnessed fluctuating water levels, further impacting tourism. Low water levels around docks and islands on Lake St. Lawrence has made boating activities impossible ("What one region's water level woes reveal about climate change and the St. Lawrence river," 2022). This region is known for its fishing, beaches, and boating, and the decline in water levels threatens both tourism and local life. According to public data from the International Lake Ontario-St. Lawrence River Board, the Lake St. Lawrence dropped below the lowest minimums ever recorded multiple times over the last few years, in May 2020, May 2021, and July, August, September and October 2022 ("International lake Ontario-St. Lawrence river board," n.d.).

Wildlife and nature-based tourism experiences have also been affected by climate change. Changing weather patterns have disrupted bird migrations in the Gaspésie region, impacting birdwatching enthusiasts and tour operators ("Bird watching in Gaspésie," 2023). Furthermore, warmer waters in the St. Lawrence River have attracted new species and changed the behaviour of the whales, altering the dynamics of whale watching. The presence of humpback whales has increased, while some traditional species like the beluga have become less predictable in their appearances. Tour operators in Tadoussac have had to adapt to these changes by diversifying their offerings and educating visitors about the evolving marine ecosystem (Chion et al., 2019).

Adaptation Strategies

Quebec's tourism industry is proactively responding to the challenges posed by climate change with a range of strategic measures. To combat the impact of warming temperatures on winter sports, ski resorts in the province have been investing in snow machines and snow production ahead of peak periods like spring break. Additionally, efforts are being made to reduce, reuse, and recycle, although there's room for improvement in this area. ("Warm weather and rain dampen winter sport enthusiasm," 2023). These combined efforts are bringing them closer to consistent snow cover throughout the ski season.

Various government-funded initiatives in Quebec are addressing climate change adaptation. The "Sous les pavés" project in Montreal transforms asphalted areas into green, sustainable spaces, mitigating heat impact and improving rainwater management. Since its inception, the project has facilitated the removal of nearly 3000 m2 of asphalt, the planting of over 300 trees, and the sowing of nearly 3,500 plants ("Le CEUM," n.d.). Other cities like Thurso, characterized by a large number of heat islands located mainly near the residential sector of the town, are exploring solutions to reduce mineralization in their downtown areas to reduce health risks. Coastal towns like Sainte-Flavie are taking steps to protect against rising water levels and flooding by replenishing beaches. And in northern Quebec, climate change's impact on permafrost is being addressed through research and knowledge development by the Chaire de recherche en partenariat sur le pergélisol au Nunavik at Université Laval ("Adapting to climate change," n.d.). In addition to these large scale projects, Infrastructure Quebec has reported that there are over 130 ongoing or completed projects in Quebec focusing on culture and recreation infrastructure, such as the creation and upkeep of community pools, parks, activities, and more ("Housing and infrastructure project map," 2023).

Furthermore, Tourisme Montréal has joined the World Council on Sustainable Tourism, emphasizing environmentally responsible practices, training, and eco-responsible guidelines to make the city a leading sustainable tourist destination by 2030. The province also has a Sustainable Tourism Action Plan, focusing on greener tourism, respect for host communities, and local sourcing ("Quebec embarks on a shift toward sustainable tourism," 2023). As noted earlier, smaller ski towns are being disproportionately impacted by the effects of climate change on their main source of income. Because of this, they are investing in new economic developments to diversify their tourism options, such as building new restaurants, attractions, and hotels ("Quebec embarks on a shift toward sustainable tourism," 2023). With all of this in mind, Quebec was ranked first in North America for its environmentally responsible practices, according to the Global Destination Sustainability Index from November 2022 ("Quebec embarks on a shift toward sustainable tourism," 2023).

The province recently published their three year sustainable tourism action plan (2022-2024), which includes actions and goals that aim to improve quality of life for residents, help businesses in the Québec City region transition to sustainable and responsible tourism, and promote sustainable and responsible tourism products and initiatives in the Québec City region (Quebec Cité, 2022). Moreover, policies such as the Sustainable Forest Development Act, the Act respecting the conservation of wetlands and bodies of water, and the Act respecting the conservation and development of wildlife safeguard natural resources, are vital for both tourism and habitat preservation. These policies play a pivotal role in ensuring the longevity of the tourism industry while protecting the environment.

Conclusion

In conclusion, this paper has illustrated the wide-ranging impacts of climate change on Quebec's tourism industry, highlighting challenges such as the disruption of outdoor activities, damage to infrastructure, shifts in wildlife patterns, and economic repercussions. Winter tourism, particularly skiing, faces serious threats from shorter snow seasons and warmer temperatures, while regions dependent on nature-based and coastal tourism grapple with environmental degradation like permafrost thawing and rising sea levels. The changing behaviour of wildlife, such as the unpredictability of whale and bird migrations, further underscores the profound effects of climate change on Quebec’s tourism experience.

Despite these challenges, Quebec has made notable strides in implementing adaptation strategies. Initiatives like snow production for ski resorts, sustainable urban projects like "Sous les pavés," and comprehensive government policies on resource management and sustainable tourism are steps in the right direction. Tourisme Montréal’s leadership in sustainable tourism and the province’s three-year sustainable action plan showcase a commitment to long-term resilience. However, while these efforts are commendable, more robust and widespread adaptation is needed, particularly in regions heavily reliant on climate-sensitive tourism, like the northern and coastal areas.

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